

Port Interfacing In Java Through Microcontroller



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A serial port interfacing circuit built around AT89C51 microcontroller is presented here for testing and understanding a serial port emulator. It can be used to control a USB port also by using a serial to USB converter.

This project can be used to control a device through the serial COM port or USB port of a PC. You can send data to the device from the PC and then receive it back on the PC through the serial port. Here we describe control of eight LEDs from the PC. The LEDs are connected across the output port of the microcontroller. The user interface program is written in Java.

Circuit description

Port P1 (refer Fig. 1) of the microcontroller is the input port and P0 the output port. The input is given through

input switches S1 through S8. The corresponding output is indicated on the LEDs (LED1 through LED8). The microcontroller is synchronised with the software at a speed of 9.6 kbps. For this, a crystal of 11.0592 MHz frequency is used. This value can be set at either the AT89C51 or by the software easily and accurately.

The microcontroller has TTL input/output (I/O) logic signal level, whereas a computer has RS232 I/O. The TTL logic works with 0V as 'low' and +5V as 'high'. In RS232 protocol, 'high' is represented by the range -3V to -25V and 'low' by +3V to +25V. The range -3V to +3V is undefined.

So the output signals available through

RS232 from the PC are not compatible with the signal requirements of AT89C51. Therefore we have used a MAX232, an RS232-to-TTL converter. Thus serial data communication can be done efficiently.

Power supply

The power supply section is shown in Fig. 2. The circuit is powered by 5V

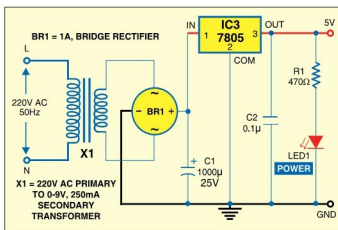


Fig. 2: Power supply circuit

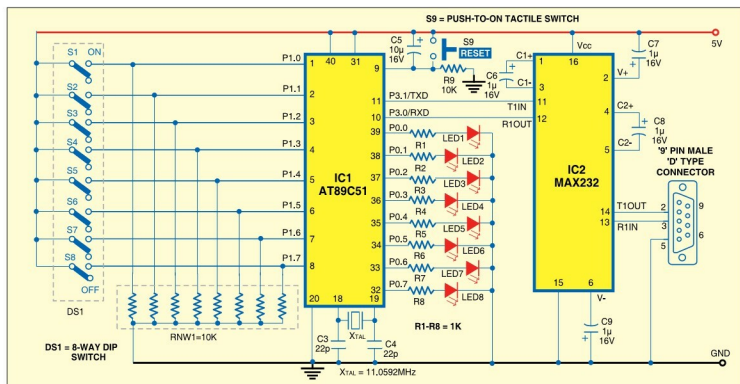


Fig. 1: Circuit for serial port interfacing with LEDs in Java through microcontroller